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Psychometric Properties of the GAD-7 among College Students: Reliability, Validity, Factor Structure, and Measurement Invariance

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Abstract

The GAD-7 is a brief, freely available questionnaire measuring anxiety, which is a common mental health problem among college students. The current study examined the psychometric properties of the GAD-7 among undergraduate students from a public midwestern university (n=582; M=19.0±1.0 years-old; 79.4% women; 81.6% White) who completed the GAD-7, State-Trait Anxiety Inventory-Trait subscale (STAI-T), and Behavioral Inhibition/Activation Scale-Reward subscale (BIAS-R). The psychometric properties of two GAD-7 models were examined: a one-factor model, equivalent to a single total score, and a two-factor model, including cognitive-emotional and somatic tension factors. The GAD-7 total score had excellent reliability (both α and ω=0.91) and evidence for convergent (r with STAI-T=0.70, p<.001) and discriminant validity (r with BIAS-R=−0.04, p=.350). The cognitive-emotional factor had excellent reliability (α=0.90; ω=0.91), whereas the somatic tension factor had adequate reliability (α=0.76; ω=0.77). Both factors had evidence for convergent validity (cognitive-emotional: r with STAT-T=0.70, p<.001; somatic tension: r with STAT-T=0.60, p<.001) and discriminant validity (cognitive-emotional: r with BIAS-R=−0.05, p=.227; somatic tension: r with BIAS-R=−0.02, p=.700). Confirmatory factor analyses supported both the one-factor model (RMSEA=.098, CFI=.994) and the two-factor model (RMSEA=.069, CFI=.997), albeit with a high inter-factor correlation (r=.91). Measurement invariance was supported across gender, race/ethnicity, and college class for both models. The findings support using the GAD-7 total score among U.S. college students, as is common in practice, but also provided evidence for the two-factor model in this population. Considering its parsimony, the total score would be preferable in applied settings for a quick measurement of anxiety symptom severity.

Keywords

GAD-7; Anxiety; Young Adult; Gender Identity; Racial Groups; Psychometrics; Factor Analysis; Statistical

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Declaration of Interest Statement
The authors have no conflicts of interests to disclose.

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Introduction

Mental health disorders typically emerge in early adulthood (De Lijster et al., 2016) and rates can be especially high in college students (Eisenberg et al., 2013). A survey of 571 college campus counseling centers found that anxiety was the number one reason that students sought counseling (LiViness et al., 2017). Anxiety disorders are the most common mental health conditions reported among college students at 11.9% (Pedrelli et. al, 2015). Many college students likely experience inflated rates of anxiety, related to multiple aspects of collegiate life, such as increased responsibility, financial concerns, academic performance, and acceptance to graduate programs, with rates of anxiety and other disorders (e.g., depression) varying across discipline and class rank (Bieter et al., 2015; Lipson et al., 2016). Identifying students' mental health disorders can result in earlier intervention and improve their chances of degree completion (Hunt et al., 2010).

For students with anxiety, using a self-report questionnaire may be the first step in identifying a problem and encouraging them to seek treatment. Self-report questionnaires allow for both professionals and nonprofessionals to identify a probable disorder and assess the severity of symptoms. Ideally, such questionnaires are brief, accurate, and easy to administer. Many are available online and can be done quickly, privately, and remotely. The GAD-7 is a seven-item questionnaire used as a screening tool for generalized anxiety disorder (GAD) and as a measure for anxiety symptom severity. The GAD-7 began as an anxiety screener for primary care but can be used to measure anxiety symptoms in a variety of individuals including inpatient psychiatric patients (Johnson et al., 2019; Beard & Bjorgvinsson, 2014), infertility patients (Omani-Samani et al., 2018), Veterans (Shepardson et al., 2019), college students (Fu et al., 2020; Byrd-Bredbenner et al., 2020), and the general population (Lowe et al., 2008). Prior psychometric research has shown the GAD-7 to have excellent internal consistency (α=0.92), good test-retest reliability (r=0.83), construct validity, and good sensitivity/specificity when detecting GAD (Spitzer, 2006). The GAD-7 was developed to produce a single score, aligning with a one-factor model, but a two-factor model has also been proposed (Kertz et al., 2012). The one-factor model involves all seven items loading onto a single factor of generalized anxiety, whereas the two-factor model splits anxiety symptoms into two correlated constructs: cognitive-emotional (e.g., nervousness, worry, and fear) and somatic tension (i.e., e.g., restlessness and irritability).

Psychometric evidence within a specific population of interest (e.g., college students) represents an important contribution of psychological science to assessment practice. When clinicians and researchers appreciate the reliability and validity of a test score within a specific population, they have a stronger empirical foundation to support the use of a psychological test or questionnaire with an individual from that population. Whereas some studies, such as those in Korea (Lee & Kim, 2019), Saudi Arabia (Alghadir et al., 2020) and Portugal (Bartolo et al., 2017), have examined the psychometric properties of the GAD-7 in college students, few psychometric studies have involved college students residing in the United States (e.g., Byrd-Bredbenner et al., 2020, 2021). Even fewer have taken demographic factors such as race/ethnicity and gender into account within this population (Sriken et. al, 2021). Women are typically twice as likely to be diagnosed with GAD (Vesga-Lopez et. al, 2008) and the prevalence of anxiety disorders varies by race and

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ethnicity, with White Americans being more likely to meet criteria for anxiety disorders than Asian, Hispanic, and African Americans (Asnaani et al., 2010). The objective of the current study was to expand on prior research on the psychometric properties of the GAD-7 in an undergraduate sample from an American university, assessing the reliability and convergent/discriminant validity of its scores, its factor structure, and measurement invariance across gender, racial/ethnic identity, and college class (i.e., freshmen vs. other classes). Investigation into the reliability, validity, and factor structure of the GAD-7 will further establish the utility of this questionnaire in clinical practice and research with college students.

Methods

Participants

Participants were recruited from students enrolled in three lower-level undergraduate psychology courses (i.e., Introduction to Psychology, Experimental Psychology, and Statistics in Psychology) at a large public midwestern university in the United States with a student body of approximately 30,000 students. Participants completed an online survey that screened for eligibility in departmental research (N=666), with participants excluded if they reported (a) prior neurological conditions (n=11), (b) neurodevelopmental disorders (n=6), and/or (c) a traumatic brain injury other than concussion (n=3); or responded incorrectly to either of two embedded validity questions (n=64) (i.e., attention checks to ensure the validity of responses). The final sample consisted of 582 undergraduate students. The sample consisted of freshmen (58.5%), sophomores (26.3%), juniors (13.2%), and seniors (1.7%). The average age of participants was 19.0 years (SD=1.0, range: 18 to 22), of whom 79.4% identified as women, 20.3% identified as men, and 0.4% identified as non-conforming/non-binary or provided no response. The sample was predominantly White (81.6%) but included representation of Black/African American (7.0%), Hispanic/Latinx (6.0%), Multiracial (4.6%), Asian (4.5%), and Native American or Pacific Islander (0.5%) college students (N.B., 2.7% reported another race/ethnicity or provided no response). The data that support the findings of this study are available from the corresponding author, upon reasonable request. The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

Measures

Participants completed a series of questionnaires including the GAD-7, State-Trait Anxiety Inventory (STAI), and the Behavioral Inhibition and Activation Scales (BIAS). Aligning with recommended approaches for detecting careless responding (Meade & Craig, 2012), embedded within the questionnaires were two validity questions, directing participants to provide a specific response to gauge if they were attending to item content.

GAD-7. The GAD-7 is a self-report questionnaire used to measure anxiety. The seven items inquire how often participants experienced anxiety-related symptoms in the past two weeks. Participants rated their symptoms on a scale from 0 (*Not at all*) to 3 (*Nearly every day*). The ratings for each score are summed to a total symptom severity score ranging from 0 to 21. This method of measuring generalized anxiety has evidence for good reliability and validity

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(Alghadir et al., 2020; Beard & Bjorgvinsson, 2014; Byrd-Bredbenner et al., 2021; Johnson, et al., 2019; Lee & Kim, 2019). Prior research has indicated that a cutoff score of ≥10 has acceptable sensitivity and specificity for detecting GAD in the general population (Ahn et al., 2019; Hinz et al., 2017; Lowe et al., 2008).

STAI. The STAI is a self-report questionnaire developed to measure state and trait anxiety (Spielberger, 1983). State anxiety refers to one's feelings of tension and apprehension that generally involves autonomic arousal. Trait anxiety represents the individual's tendency to respond to threatening stimuli with state anxiety. Form Y of the STAI consists of 40 total items, 20 each for state and trait anxiety. Items are rated on a four-point scale with some items reverse scored. The items are summed to create State and Trait anxiety scores. The Trait subscale (STAI-T) was used in the current study for assessing convergent validity, with higher scores indicating greater anxiety.

BIAS. The BIAS is based on the theory that motivation is driven by two underlying systems: behavioral inhibition and behavioral activation (Carver, 1994). The BAS (behavior activation system) is considered to regulate appetite motives where the objective is to move towards a desired goal. The BIS (behavior inhibition system) is theorized to regulate avoidance in which moving away from aversive stimuli is the goal. Subscales for the BIAS include BAS Drive, BAS Fun-Seeking, BAS Reward Response, and BIS. The BIAS was developed to identify one's sensitivity in each system. The 24 items are ranked by the participant on a scale from 1 (very true for me) to 4 (very false for me) and includes four filler questions. Respective items represent each subscale, and the scores were summed for each category, with some items reverse-scored in alignment with the original scale development. Due to largely differing constructs between the GAD-7 and the BAS Reward Response, this subscale was used to assess discriminant validity.

Procedure

A screener for participation in department research was advertised to psychology students enrolled in undergraduate psychology courses in Spring 2021 during the COVID-19 pandemic. Participants were offered course credit for completing this screener, which was administered online via the Sona Systems® participant pool management software. Only participants who provided informed consent for the use of their data in research were included in this study. This study protocol received ethical approval from a university institutional review board.

Statistical Analysis

All statistical analyses were conducted using RStudio (RStudio Team, 2020). Reliability was evaluated by calculating alpha (α) (Cronbach, 1951) and omega (ω) (McDonald, 1999). The Gilmer-Feldt composite reliability (CR_{GF}) was calculated as a supplementary estimate of reliability (Gilmer & Feldt, 1983). Convergent validity was evaluated by examining the correlations of the GAD-7 scores with the STAI-T, anticipating large correlations; and discriminant validity was evaluated by examining the correlations of the GAD-7 scores with the BIAS Reward subscale, anticipating very small correlations. A comparison of the convergent and discriminant validity coefficients was conducted (Meng et al., 1992), with

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correlations between the GAD-7 scores and the STAI-T hypothesized to be significantly larger than the correlations with the BIAS Reward subscale.

Confirmatory factor analysis. A confirmatory factor analysis was conducted in R using Lavaan (Rosseel, 2012). Both one-factor and two-factor models were specified, estimated using weighted least squares with mean and variance adjusted (WLSMV), and evaluated for model fit. The sample size was sufficient for confirmatory factor analysis of polytomous data using robust weighted least squares (Moshagen & Musch, 2014). Consistent with the GAD-7 total sum score used in practice, the one-factor model proposes all items loading onto a single latent factor of generalized anxiety. As an alternative model investigated in prior research on the GAD-7 (Kertz et al., 2012), the two-factor model assigns cognitive-emotional symptoms (items 1, 2, 3, 7) and somatic tension symptoms (items 4, 5, 6) on separate factors. To interpret model fit, multiple fit indices were calculated. The chi-square test of model fit was conducted, but is sensitive to large sample size, with increased power to reject the null hypothesis (i.e., indicating poor fit) (Tanaka, 1987). In turn, alternative fit indices were interpreted, including the Root Mean Square Error of Approximation (RMSEA), for which a value <.08 is indicative of acceptable fit (Browne & Cudeck, 1993); the Comparative Fit Index (CFI) and Tucker-Lewis Index (TLI), for which values >.95 are indicative of excellent fit (Hu & Bentler, 1999); and the Standardized Root Mean Squared Residual (SRMR) for which a value of <.08 is indicative of good fit (Hu & Bentler, 1999).

Invariance analysis and group comparisons. Measurement invariance was determined via a multistep approach that compares nested models with added constraints: configural, weak, and strong (Bontempo et al., 2012). The configural model holds the same factor structure across groups. The weak model constrains the factor loadings to be equal across groups. The strong model constrains thresholds to be equal across groups when using ordinal data (Finney & DiStefano, 2013; Millsap & Yun-Tein, 2004). If demonstrated, strong invariance indicates that the means, while not equal, are meaningfully comparable between groups and differences in means may be attributed to differences in the construct rather than issues with the measure across groups. Measurement invariance was tested across gender [men (n=119) and women (n=463)], race/ethnicity [White (n=455) and other racial/ethnic identities (n=119)], and college class [freshman (n=348) and other classes (n=234)], for both the one-factor and two-factor models using the Lavaan package in R Studio (Rosseel, 2012). For some groupings, the sample was not evenly distributed. That said, prior simulation studies with sample sizes as low as 100 had sufficient power for confirmatory factor analyses with similar numbers of factors, numbers of indicators, and magnitudes of loadings (Wolf et al., 2013); and ΔCFI was not affected by sample size in invariance analysis simulations, with samples of 150 performing similarly to samples of 300 (Cheung & Rensvold, 2009).

To establish measurement invariance, a change in fit between the compared models must be either non-significant per a $\Delta\chi^2$ test, with modifications when using WLSMV (Bryant & Satorra, 2012; Satorra, 2000) or show no more than −.01 decline in CFI (Cheung & Rensvold, 2002) or −0.015 decline in ΔRMSEA (Chen, 2007). With the establishment of measurement invariance, participants were compared via t-tests on the GAD-7 total score

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and subscale scores (i.e., cognitive-emotional and somatic tension symptoms) based on gender and racial/ethnic identity.

Results

The sample (n=582) had the following average ratings for the administered questionnaires: GAD-7 (M=7.4, SD=5.7, range: 0 to 21), STAI-T (M=43.2, SD=10.6, range: 21 to 72), and BIAS Reward (M=7.4, SD=2.0, range: 5 to 15). In terms of cutoffs for the GAD-7 (Spitzer et al., 2006), 52.1% exceed the cutoff for mild anxiety (GAD-7≥5), 33.0% exceeded the cutoff for moderate anxiety (GAD-7≥10), and 12.0% exceeded the cutoff for severe anxiety (GAD-7≥15). Participants endorsing at least one item on the GAD-7 reported how difficult these symptoms made their work, homecare, or social life, of whom 29.3% responded *not difficult at all*, 51.6% responded *somewhat difficult*, 12.0% responded *very difficult*, and 7.1% responded *extremely difficult*. Among participants exceeding the provisional diagnostic cutoff (i.e., GAD-7≥10), 97.4% responded that the symptoms made their life *somewhat difficult*, *very difficult*, or *extremely difficult,* meaning 32.1% of the full sample (n=187) endorsed GAD symptoms above the recommended clinical cutoff for diagnosis and reported cooccurring functional impairment.

Reliability and Validity

Reliability of the GAD-7 total score, with a 95% confidence interval, were estimated using both alpha (α=0.91 [.90, .92]) and omega (ω=0.91 [.90, .92]), with both producing equivalent values indicating excellent reliability of the total GAD-7 sum score. For the two-factor model, the reliability of the cognitive-emotional factor was excellent (α=0.90 [.89, .91]; ω=0.91 [.90, .92]) and the reliability of the somatic tension factor was adequate (α=0.76 [.72, .79]; ω=0.77 [.74, .81]). As a supplemental estimate of reliability, the CR_{GF} reliability coefficient was calculated for the total score (CR_{GF}=.91), cognitive-emotional score (CR_{GF}=.91), and somatic tension score (CR_{GF}=.77), producing essentially identical values. The correlation between the GAD-7 and STAI-T was examined to assess convergent validity, which was significant and of large magnitude (r=0.70 [.65, .74], p<.001), and the correlation between the BIAS Reward subscale and GAD-7 was non-significant and of extremely small magnitude (r=−0.04 [−.12, .04], p=.350). The correlation between the GAD-7 and STAI-T was significantly larger than the correlation between the GAD-7 and BIAS Reward subscale (z=15.80, p<.001). The same analyses were conducted for the two-factor model, with moderate to high correlations between the STAI-T and the cognitive-emotional (r=0.70 [.65, .74], p<.001) and somatic tension factors (r=0.60 [.54, .65], p<.001) and low correlations between the BIAS Reward subscale and the cognitive-emotional (r=−0.05 [−.13, .03], p=.227) and somatic tension factors (r=−0.02 [−.10, .07], p=.700). The correlations between the STAI-T and the two GAD-7 factors were significantly higher than the correlations between the BIAS Reward subscale and the two GAD-7 factors (i.e., cognitive-emotional: z=13.11, p<.001; somatic tension: z=8.74, p<.001).

Confirmatory Factor Analysis

Confirmatory factor analysis evaluated both one-factor and two-factor structures of the GAD-7. The one-factor model resulted in all items loading onto a generalized anxiety

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latent factor. The model had excellent fit per the CFI and TLI (i.e., CFI=0.994, TLI=0.992, SRMR=.036), but did not meet the conventional threshold for acceptable fit based on the RMSEA (i.e., RMSEA=0.098, $p<0.05$). All loadings were significant on the general factor (λ range: 0.63 to 0.97, $p<0.001$). All loadings and residual values are presented within a factor diagram provided in Figure 1. The two-factor model included cognitive-emotional symptoms (items 1, 2, 3, 7) and somatic tension symptoms (items 4, 5, 6) on separate factors. The two-factor model also had excellent fit per the CFI, TLI, and SRMR (CFI=0.997, TLI=0.996, SRMR=.028) and passed the threshold for acceptable fit for the RMSEA (RMSEA=0.069, $p<0.05$). However, the correlation between the two factors was very high ($r=0.91$). Loadings are reported in a factor diagram provided in Figure 2.

Measurement Invariance and Group Comparisons

The results of measurement invariance analyses are presented in Table 1 for the one-factor model and Table 2 for the two-factor model. Configural, weak, and strong invariance were established across gender, race/ethnicity, and college class. For the one-factor and two-factor models, there was no change in CFI between the configural and strong models, and all $\Delta\chi^2$ values were non-significant, meaning model fit was comparable across all models, even with added constraints. There was one notable loss of invariance per the ΔRMSEA from the configural to weak invariance models for gender (i.e., ΔRMSEA=−.032), but fit was improved with the added constraints of item thresholds for the strong model.

With the establishment of measurement invariance, groups were compared on the total score and the two subscales of cognitive-emotional and somatic tension symptoms. The results of these group comparisons are reported in Table 3. A moderate difference was found between gender groups with women reporting higher symptom severity than men on the total score, cognitive-emotional score, and somatic tension score. There were no significant differences between White participants and participants of other racial/ethnic identities on the total score or either subscale. Similarly, there was no significant difference found between freshman students and other classes.

Discussion

The present study evaluated the psychometric properties of the GAD-7 in a sample of college students. The results indicated excellent reliability along with discriminant and convergent validity for the GAD-7 total score. A confirmatory factor analysis was conducted to assess the construct validity of both a one-factor model, reflecting the total sum score often used in research and practice, and two-factor model of the GAD-7, dividing symptoms across cognitive-emotional and somatic tension symptom factors. The results of the factor analysis supported the single-factor structure, with strong loadings for all items onto this factor and acceptable fit based on some (i.e., CFI=0.994, TLI=0.992), but not all metrics (RMSEA=0.098).

The two-factor model has been proposed and implemented in previous psychometric studies (Baratolo et al., 2017; Beard et al., 2014; Kertz et al., 2012; Johnson et al.; 2019). The two-factor model showed excellent reliability for the cognitive-emotional factor and adequate reliability for the somatic tension factor, with both factors showing evidence for convergent

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and discriminant validity. While the two-factor structure met the threshold for good fit based on all indices, the minor increase in fit over the one-factor model, along with the high correlation between the somatic tension and cognitive-emotional latent variables (i.e., r=.91, consistent with prior research, Baratolo et al., 2017), favors the parsimonious one-factor structure as well.

Measurement invariance analyses indicated consistent model fit across gender, college class, and racial/ethnic identity for both the one-factor and two-factor models. There was one notable difference in the measurement invariance findings across gender, where the ΔRMSEA indicated a loss of invariance when comparing the weak model to the configural model. This finding indicates the relationship between the cognitive-emotional and somatic tension factors and the GAD-7 items differed across groups, albeit the additional constraint of the item thresholds for the strong model showed an improvement in fit and overall excellent fit (i.e., RMSEA=.000). As observed in prior research (Garcia-Barrera et al., 2013), fit indices can improve with additional constraints on parameters introduced in the strong model. This finding was based on a single fit index, which was discrepant from the ΔCFI and $\Delta\chi^2$ results. For both models, χ^2 difference tests were non-significant from the configural to the strong models, indicating that means for the GAD-7 total score and subscale scores (i.e., cognitive-emotional and somatic tension) can be meaningfully compared between men and women, between freshmen and other classes, and between White college students and college students of other racial/ethnic identities. When compared in mean symptom ratings, women reported higher anxiety symptoms than men, but there were no differences between college classes or racial/ethnic groups.

The current findings are in alignment with similar studies using college students and other populations. The reliability estimates for the current study (both α and ω=0.91) were very similar to, or even equivalent to, those found in other samples of college students in the United States (α=0.92; Byrd-Bredbenner, 2021), Korea (α=0.91; Lee et al., 2019), and Saudi Arabia (α=0.83; Alghadir et al., 2020), indicating that the GAD-7 may be a reliable method of measuring anxiety in college students of various cultural backgrounds. Internationally, clinical and general population studies reported similar reliability estimates to those found in this study. Both inpatient and outpatient participants with psychiatric conditions undergoing various treatments in the United States and Norway showed good-to-excellent reliability of the GAD-7 at pre-treatment (α=0.83; α=0.87) and post-treatment (α=0.86; α=0.91) assessments (Beard & Bjorgvinsson, 2014; Johnson et al., 2019). Studies on general population samples in Germany (α=.85; Hinz et al., 2017) and Korea (α=.85; Ahn et al., 2019) showed good-to-excellent reliability as well. Convergent validity results aligned with a Japanese study that also correlated the GAD-7 with the STAI (r=0.74; Doi et al., 2018), finding a comparable convergent validity coefficient to the current study (i.e., r=.70). Results for the CFA support previous findings that the GAD-7, as a single-factor model, meets most requirements for good fit (Byrd-Bredbenner et al., 2021; Ryan et al., 2013).

For the purposes of briefly assessing symptom severity, a single GAD-7 total score is a reliable tool among college students. Future studies regarding the factor structure of the GAD-7 might investigate whether different symptom dimensions of anxiety (i.e.,

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cognitive-emotional versus somatic tension) may inform clinical interventions among college students. Conceptually, students with higher symptoms of somatic tension may benefit from behavioral interventions, whereas participants with greater cognitive-emotional symptoms may benefit from cognitive treatment strategies. Collectively, these findings demonstrate the broad utility of the GAD-7 total score, which has consistent evidence for reliability and validity across multiple populations, but also supports further exploration on the utility of the two-factor model in research and practice.

The current study has limitations. For example, the sample consisted of college students from a psychology department at a single public university in the United States. Although participants were recruited from psychology courses, about two-thirds of the students typically enrolled in these courses report having majors other than psychology, meaning these findings could apply to students studying disciplines other than psychology. That said, the sample was 79.4% women, 81.6% White, and 59.8% freshman class. These characteristics may interfere with the generalizability of the results to other populations, including among college students of different ages; however, the psychometric properties of the GAD-7 have been relatively comparable across different populations (per prior research across varied samples, α range: 0.83 to 0.92; Ahn et al., 2019; Alghadir et al., 2020; Beard & Bjorgvinsson, 2014; Byrd-Bredbenner et al., 2021; Hinz et al., 2017; Johnson et al., 2019; Lee et al., 2019; Ryan et al., 2013).

Information about socioeconomic status (e.g., family income) was not collected, which may relate to anxiety severity in college students. Significant relationships have been observed between socioeconomic status and testing anxiety, academic self-efficacy, and achievement (Akanbi, 2013; Bozkurt et al., 2021). All measures in this study were administered online. In at least one instance, the GAD-7 has shown slightly higher reliability when administered face-to-face (α=.89) compared to telephone administration (α=.82) (Ryan et al., 2013). In-person interviews conducted by a trained mental health professional is often an ideal way of gathering data of this nature but is more time-intensive and costly. The results were based on only self-report measures that were collected at a single timepoint in a cross-sectional design. Prospective studies may consider using a longitudinal design to examine the reliability and validity of the GAD-7 over time and over the course of treatment in outpatient college counseling settings.

Anxiety rates are high among college students and can become chronic without proper treatment. A reliable and accessible screener for anxiety is crucial for timely detection and intervention (Bieter et al., 2015; Hunt et al., 2010; Lipson et al.; 2015). The GAD-7 is a brief, simple, and publicly available questionnaire, making it a feasible method for screening for anxiety in many populations. The current results support the reliability and validity of the GAD-7 for measuring anxiety in college students in the United States. The one-factor model provides evidence of construct validity for the total score often applied in research and clinical practice. Previous literature supports the use of the GAD-7 in populations other than students and U.S. residents, suggesting reliability and validity amongst a diversity of individuals. The GAD-7 may be useful in a college counseling setting as a no-cost and quickly administered screening instrument, and future studies could focus on testing the GAD-7 as a screening tool for generalized anxiety in a college counseling practice.

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Specifically, it may be value to determine if the current threshold for the GAD-7 (i.e., ≥10) would indicate need for further assessment or intervention among college students, with notably 33.0% of this sample meeting or exceeding that threshold. Such a follow-up study could help identify individuals who may require additional evaluation or psychological treatment based on GAD-7 scores.

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References

Ahn J, Kim Y, Choi K (2019). The psychometric properties and clinical utility of the Korean version of GAD-7 and GAD-2. Frontiers in Psychology, 10(127). 10.3389/fpsyt.2019.00127

Akanbi ST (2013). Comparisons of test anxiety level of senior secondary school students across gender, year of study, school type and parental educational background. Ife Psychologia, 21(1), 40–54.

Alghadir A, Manzar MD, Anwer S, Albougami A (2020). Psychometric properties of the generalized anxiety disorder scale among Saudi university male students. Neuropsychiatric Disease and Treatment, (16), 1427–1432. 10.2147/NDT.S246526 [PubMed: 32606696]

Asnaani A, Richie A, Dimaite R, Hinton D, Hofmann S (2010). A cross-ethnic comparison of lifetime prevalence rates of anxiety disorders. The Journal of Nervous and Mental Disease, 198(8). 10.1097/NMD.0b013e3181ea169f

Bartolo A, Montiero S, Periera A (2017). Factor structure and construct validity of the Generalized Anxiety Disorder 7-item (GAD-7) among Portuguese college students. Methodological Issues, 33(9). 10.1590/0102-311X00212716

Beard C, & Bjorgvinsson T (2014). Beyond generalized anxiety disorder: Psychometric properties of the GAD-7 in a heterogeneous psychiatric sample. Journal of Anxiety Disorders. 28(6), 547–552. 10.1016/j.janxdis.2014.06.002 [PubMed: 24983795]

Bieter R, Nash R, McCrady M, Rhoades D, Linscomb M, Clarahan M, Sammut S (2015). The prevalence and correlates of depression, anxiety, and stress in a sample of college students. Journal of Affective Disorders, (173), 90–96. 10.1016/j.jad.2014.10.054

Bontempo DE, Grouzet FME, & Hofer SM (2012). Measurement issues in the analysis of within-person change. In Longitudinal data analysis: A practical guide for researchers in aging, health, and social sciences (pp. 97–142). Routledge/Taylor & Francis Group.

Bozkurt S, Çoban Ö, Özdemir M, & Özdemir N (2021). How Leadership, School Culture, Collective Efficacy, Academic Self-Efficacy, and Socioeconomic Status Affect Student Achievement. Education and Science, 46(207), 1–18. 10.15390/EB.2021.9338

Browne MW, & Cudeck R (1993). Alternative ways of assessing model fit. Sociological Methods and Research, 21(2), 230–258. 10.1177/0049124192021002005

Bryant FB, & Satorra A (2012). Principles and practice of scaled difference chi-square testing. Structural Equation Modeling: A Multidisciplinary Journal, 19(3), 372–398. 10.1080/10705511.2012.687671

Byrd-Bredbenner C, Eck K, Quick V (2020). Psychometric properties of the Generalized Anxiety Disorder-7 and Generalized Anxiety Disorder mini in United States students. Frontiers in Psychology. 10.3389/fpsyg.2020.550533

Byrd-Bredbenner C, Eck K, Quick V (2021). GAD-7, GAD-2, and GAD-mini: Psychometric properties and norms of university students in the United States. General Hospital Psychiatry, 69, 61–66. 10.1016/j.genhosppsych.2021.01.002 [PubMed: 33571925]

Carver CS, & White TL (1994). Behavioral inhibition, behavioral activation, and affective responses to impending reward and punishment: The BIS/BAS Scales. Journal of Personality and Social Psychology, 67(2), 319–333. 10.1037/0022-3514.67.2.319

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Chen FF (2007). Sensitivity of goodness of fit indexes to lack of measurement invariance. Structural Equation Modeling, 14(3), 464–504. 10.1080/10705510701301834

Cheung GW, & Rensvold RB (2002). Evaluating goodness-of-fit indexes for testing measurement invariance. Structural Equation Modeling: A Multidisciplinary Journal, 9(2), 233–255. 10.1207/S15328007SEM0902_5

Cronbach L (1951). Coefficient alpha and the internal structure of tests. Psychometrika, 16, 297–334. 10.1007/BF02310555

De Lijster JM, Dierckx B, Utens E, Verhulst F, Zieldorff C, Dieleman G, Legerstee J (2017). The age of onset of anxiety disorders: A meta-analysis. The Canadian Journal of Psychology, 62(4), 237–246. 10.1177/0706743716640757

Doi S, Ito M, Takebayashi Y, Muramatsu K, Horikoshi M (2018). Factorial validity and invariance of the 7-item generalized anxiety disorder scale (GAD-7) among populations with and without self-reported psychiatric diagnostic status. Frontiers in Psychology, 9, 1741. 10.3389/fpsyg.2018.01741 [PubMed: 30283386]

Eisenberg D, Hunt J, Speer N (2013). Mental health in American colleges and universities: variation across student subgroups and across campuses. The Journal of Nervous and Mental Disease, 201(1), 60–67. 10.1097/NMD.0b013e31827ab077 [PubMed: 23274298]

Finney SJ, & DiStefano C (2013). Non-normal and categorical data in structural equation modeling. In Hancock GR & Mueller RO (Eds.), Quantitative methods in education and the behavioral sciences: Issues, research, and teaching. Structural equation modeling: A second course (p. 439–492). IAP Information Age Publishing.

Fu W, Yan S, Zong Q, Anderson-Luxford D, Song Q, Lv Z, Lv C (2021). Mental health of college students during the COVID-19 epidemic in China. Journal of Affective Disorders, 280, 7–10. 10.1016/j.jad.2020.11.032

Garcia-Barrera MA, Karr JE, & Kamphaus RW (2013). Longitudinal applications of a behavioral screener of executive functioning: Assessing factorial invariance and exploring latent growth. Psychological Assessment, 25(4), 1300–1313. 10.1037/a0034046 [PubMed: 23914956]

Gilmer JS, & Feldt LS (1983). Reliability estimation for a test with parts of unknown lengths. Psychometrika, 48(1), 99–111. 10.1007/BF02314679

Hinz A, Klein A, Brahler E, Glaesmer H, Luck T, Riedel-Heller S, Wirkner K, Hilbert A (2017). Psychometric evaluation of the Generalized Anxiety Disorder Screener GAD-7, based on a large German general population sample. Journal of Affective Disorders, 210, 338–344. 10.1016/j.jad.2016.12.012 [PubMed: 28088111]

Hu LT, & Bentler PM (1999). Cutoff criteria for fit indexes in covariance structure analysis: Conventional criteria versus new alternatives. Structural Equation Modeling, 6(1), 1–55. 10.1080/10705519909540118

Hunt J, Eisenberg D, Kilbourne A (2010) Consequences of receipt of a psychiatric diagnosis for completion of college. Psychiatric Services, 61(4), 399–404. 10.1176/ps.2010.61.4.399 [PubMed: 20360280]

Johnson S, Ulvenes P, Oktedalen T, Hoffart A (2019). Psychometric properties of the General Anxiety Disorder 7-Item (GAD-7) scale in a heterogeneous psychiatric sample. Frontiers in Psychology. 10.3389/fpsyg.2019.01713

Kertz S, Bigda-Peyton J, Bjorgvinsson T (2012). Validity of the Generalized Anxiety Disorder-7 scale in an acute psychiatric sample. Clinical Psychology & Psychotherapy, 20(5), 456–464. 10.1002/cpp.1802 [PubMed: 22593009]

Lee B, & Kim Y (2019) The psychometric properties of the Generalized Anxiety Disorder scale (GAD-7) among Korean university students. Psychiatry and Clinical Psychopharmacology, 29(4), 864–871. 10.1080/24750573.2019.1691320

LiViness P, Bershad C, Gorman K, Braun L, Murray T The Association of University and College Counseling Center Directors Annual Survey: 2019. AUCCCD. https://www.aucccd.org/assets/documents/Survey/2019%20AUCCCD%20Survey-2020-05-31-PUBLIC.pdf

Lipson S, Zhou S, Wagner B, Beck K, Eisenberg D (2015). Major differences: Variations in undergraduate and graduate student mental health and treatment utilization

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across academic disciplines. Journal of College Student Psychotherapy, 30(1), 23–41. 10.1080/87568225.2016.1105657

Lowe B, Decker O, Muller S, Brahler E, Schelberg D, Herzog W, Herzberg P (2008). Validation and standardization of the Generalized Anxiety Disorder screener (GAD-7) in the general population. Medical Care, 46(3), 266–274. 10.1097/MLR.0b013e318160d093 [PubMed: 18388841]

McDonald RP (1999). Test theory: A unified treatment. Mahwah, NJ: Lawrence Erlbaum

Meade A, & Craig BS (2012). Identifying careless responses in survey data. Psychological Methods, (3) 437–55. 10.1037/a0028085 [PubMed: 22506584]

Meng XL, Rosenthal R, & Rubin DB (1992). Comparing correlated correlation coefficients. Psychological Bulletin, 111(1), 172–175. 10.1037/0033-2909.111.1.172

Millsap RE, & Yun-Tein J (2004). Assessing factorial invariance in ordered-categorical measures. Multivariate Behavioral Research, 39(3), 479–515. 10.1207/S15327906MBR3903_4

Moshagen M, & Musch J (2014). Sample size requirements of the robust weighted least squares estimator. Methodology, 10(2), 60–70. 10.1027/1614-2241/a000068

Omani-Samani R, Maroufizadeh S, Ghaheri A, Navid B (2018). Generalized Anxiety Disorder (GAD-7) in people with infertility: A reliability and validity study. Middle East Fertility Society Journal, 23(4), 446–449. 10.1016/j.mefs.2018.01.013

Pedrelli P, Nyer M, Yeung A, Zulauf C, Wilens T (2015). College students: Mental health problems and treatment considerations. Academic Psychiatry, 39(5), 503–511. 10.1007/s40596-014-0205-9 [PubMed: 25142250]

Rosseel Y (2012). lavaan: An R Package for Structural Equation Modeling. Journal of Statistical Software, 48(2), 1–36. 10.18637/jss.v048.i02

RStudio Team (2020). RStudio: Integrated Development for R. RStudio, PBC, Boston, MA. http://www.rstudio.com/.

Ryan T, Bailey A, Fearon P, King J (2013). Factorial invariance of the Patient Health Questionnaire and Generalized Anxiety Disorder Questionnaire. British Journal of Clinical Psychology, 52(4), 438–449. 10.1111/bjc.12028 [PubMed: 24117915]

Satorra A (2000). Scaled and adjusted restricted tests in multi-sample analysis of moment structures. In Heijmans RDH, Pollock DSG, & Satorra A (Eds.), Innovations in multivariate statistical analysis: A Festschrift for Heinz Neudecker (pp. 233–247). Springer. 10.1007/978-1-4615-4603-0_17

Shepardson R, Kosiba J, Bernstein L, Funderburk J (2019). Suicide risk among Veteran primary care patients with current anxiety symptoms. Family Practice, 36(1), 91–95. 10.1093/fampra/cmy088 [PubMed: 30219849]

Spielberger CD (1989). State-Trait Anxiety Inventory: Bibliography (2nd ed.). Palo Alto, CA: Consulting Psychologists Press.

Spitzer R, Kroenke K, Williams J, Lowe B (2006). A brief measure for assessing generalized anxiety disorder: The GAD-7. Archives of Internal Medicine, 166(10), 1092–1097. 10.1001/archinte.166.10.1092 [PubMed: 16717171]

Sriken J, Johnson S, Smith H, Sherman M, Erford B (2021): Testing the factorial validity and measurement invariance of college student scores on the generalized anxiety disorder (GAD-7) scale across gender and race. Measurement and Evaluation in Counseling and Development, 55(1), 1–16. DOI: 10.1080/07481756.2021.1902239

Tanaka JS (1987). “How big is big enough?”: Sample size and goodness of fit in structural equation models with latent variables. Child Development, 58(1), 134–146. 10.2307/1130296

Vesga-Lopez O, Schneier F, Wang S, Heimberg R, Liu S, Hasin D, Blanco C (2016). Gender differences in generalized anxiety disorder: Results from the national epidemiologic survey on alcohol and related conditions (NESARC). Journal of Clinical Psychology, 69(10), 1606–1616.

Welch B (1938). The significance of the difference between two means when the population variances are unequal. Biometrika, 29(3–4), 350–362. 10.1093/biomet/29.3-4.350

Wolf EJ, Harrington KM, Clark SL, & Miller MW (2013). Sample Size Requirements for Structural Equation Models: An Evaluation of Power, Bias, and Solution Propriety. Educational and Psychological Measurement, 73(6), 913–934. 10.1177/0013164413495237

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Public Significance Statement:

The GAD-7 is a brief, freely available questionnaire for measuring generalized anxiety symptoms, developed for use in primary care settings. The current study found good reliability and validity of the GAD-7 among college students. The GAD-7 could be implemented into research and practice with college students, helping evaluate anxiety severity.

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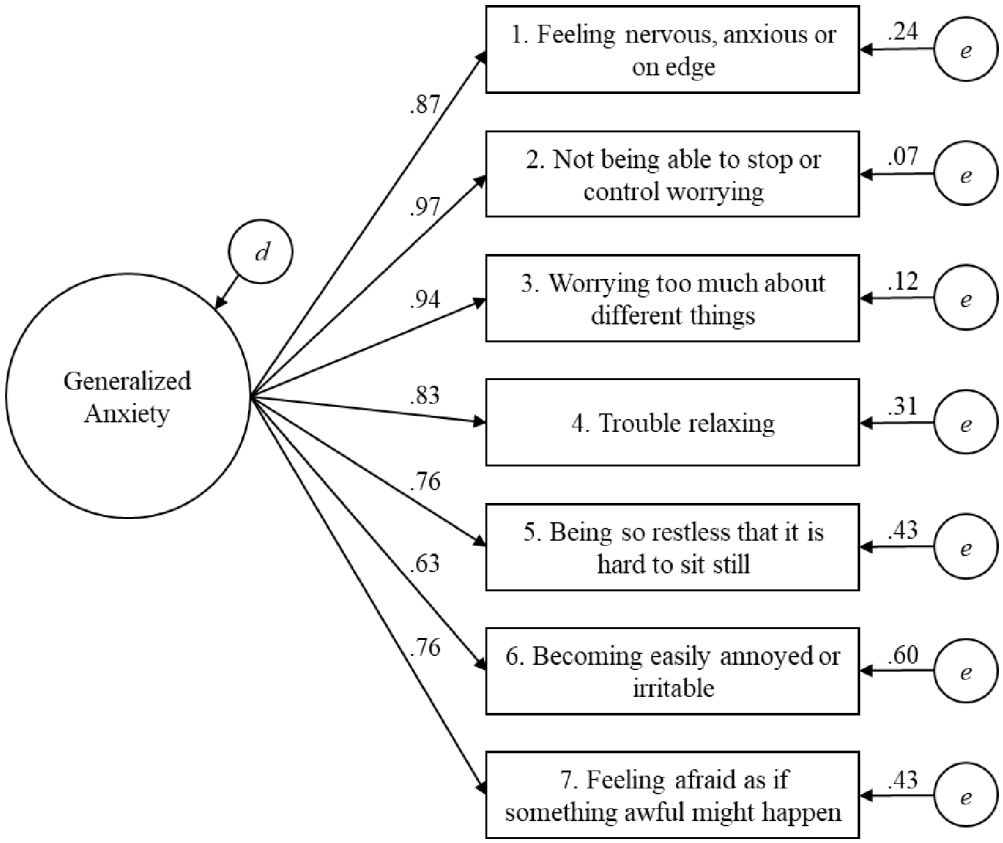


Figure 1.
One-Factor Model of the Seven-Item Generalized Anxiety Disorder Scale (GAD-7)

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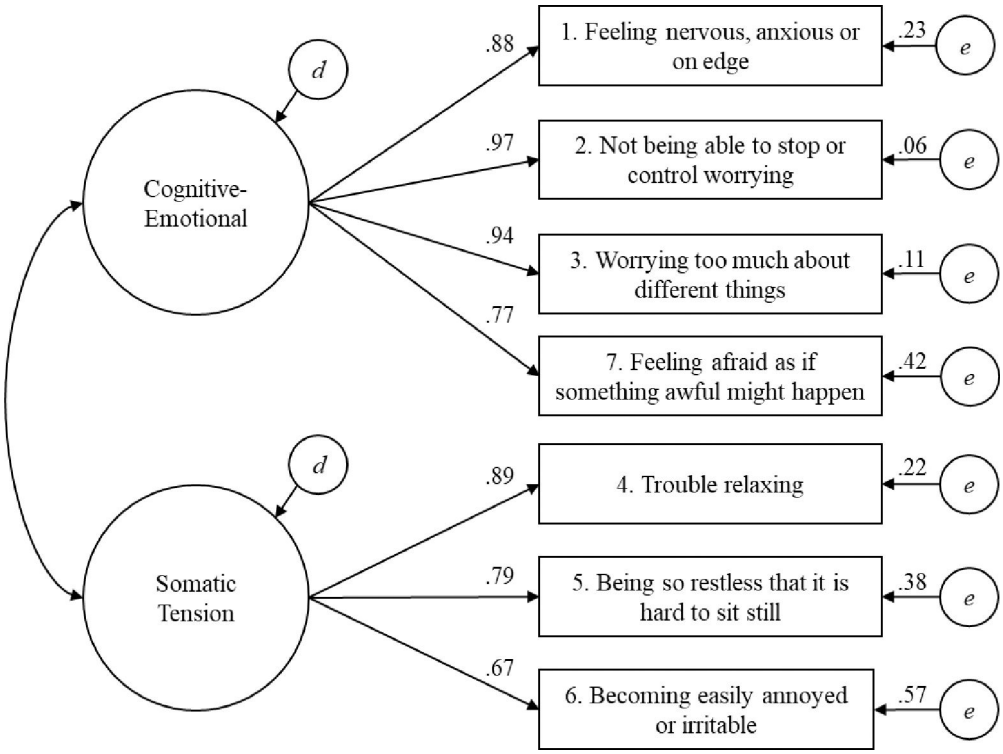


Figure 2.
Two-Factor Model of the Seven-Item Generalized Anxiety Disorder Scale (GAD-7)

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Table 1.

Measurement invariance results for the one-factor model

Group	Invariance	WLSMV χ^2	df	*p*	$\Delta\chi^2$ (*p*)	CFI	ΔCFI	TLI	SRMR	RMSEA (90% CI)	ΔRMSEA
Race/Ethnicity	Configural	53.23	28	.003	–	0.999	–	0.999	0.041	0.056 (.032–.078)	–
	Weak	62.85	34	.002	5.85 (.4402)	0.999	0.000	0.999	0.044	0.054 (.032–.075)	−0.002
	Strong	72.69	47	.010	15.95 (.2517)	0.999	0.000	0.999	0.041	0.043 (.022–.062)	−0.011
Gender	Configural	48.54	28	.009	–	0.999	–	0.999	0.039	0.050 (.025–.074)	
	Weak	63.84	34	.001	7.12 (.3103)	0.999	0.000	0.998	0.045	0.055 (.034–.076)	
	Strong	61.34	47	.078	4.39 (.9862)	0.999	0.000	0.999	0.040	0.032 (.000–.053)	
Class	Configural	49.33	28	.008	--	0.999	0.000	0.999	0.040	0.051 (.026–.074)	
	Weak	55.19	34	.012	3.82 (.7015)	0.999	0.000	0.999	0.042	0.046 (.022–.068)	
	Strong	59.97	47	.097	10.49 (.6538)	0.999	0.000	1.000	0.040	0.031 (.000–.052)	

Note. CFI=Comparative fit index; RMSEA=Root mean square error of approximation; SRMR=Standardized root square mean residuals; TLI=Tucker Lewis index; WLSMV=Weighted least squares with mean and variance adjusted. Race/ethnicity was coded as White vs. Other racial/ethnic identities, Gender was coded as Men vs. Women, and Class was coded as Freshman vs. Other classes

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Table 2.

Measurement invariance results for the two-factor model

Group	Invariance	WLSMV χ^2	df	*p*	$\Delta\chi^2$ (*p*)	CFI	ΔCFI	TLI	SRMR	RMSEA (90% CI)	ΔRMSEA
Race/Ethnicity	Configural	30.15	26	.262	–	1.000	–	1.000	0.032	0.023 (.000–.054)	
	Weak	38.33	31	.171	5.13 (.3999)	1.000	.000	1.000	0.034	0.029 (.000–.055)	
	Strong	49.31	43	.235	17.13 (.1446)	1.000	.000	1.000	0.032	0.022 (.000–.047)	
Gender	Configural	26.00	26	.463	–	1.000	–	1.000	0.031	0.000 (.000–.033)	
	Weak	40.23	31	.124	6.59 (.2533)	1.000	.000	0.999	0.036	0.032 (.000–.058)	
	Strong	38.56	43	.664	4.62 (.9696)	1.000	.000	1.000	0.032	0.000 (.000–.046)	
Class	Configural	28.89	26	.316	--	1.000	--	1.000	0.032	0.020 (.000–.052)	
	Weak	33.92	31	.329	3.49 (.624)	1.000	.000	1.000	0.034	0.018 (.000–.049)	
	Strong	37.69	43	.700	8.91 (.7107)	1.000	.000	1.000	0.033	0.000 (.000–.032)	

Note. CFI=Comparative fit index; RMSEA=Root mean square error of approximation; SRMR=Standardized root square mean residuals; TLI=Tucker Lewis index; WLSMV=Weighted least squares with mean and variance adjusted. Race/ethnicity was coded as White vs. Other racial/ethnic identities, Gender was coded as Men vs. Women, and Class was coded as Freshman vs. Other classes

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Table 3.

Group comparisons for the GAD-7 total score and subscale scores

		GAD-7 Total Score		Cognitive-Emotional		Somatic Tension	
		M	***SD***	***M***	***SD***	***M***	***SD***
Gender	Women (n=463)	8.0	5.7	5.0	3.7	3.0	2.5
	Men (n=119)	5.3	5.1	3.3	3.3	2.1	2.1
	Group Comparison	t=4.92, p<.001, d=.47		t=4.94, p<.001, d=.48		t=4.14, p<.001, d=.39	
Race/Ethnicity	white (n=455)	7.6	5.8	4.7	3.7	2.9	2.5
	Other Identities (n=127)	6.9	5.3	4.3	3.4	2.6	2.3
	Group Comparison	t=1.20, p=.115, d=.12		t=1.26, p=.105, d=.13		t=.94, p=.174, d=.09	
Class	Freshmen (n=348)	7.6	5.8	4.8	3.7	2.8	2.5
	Other Class (n=234)	7.2	5.5	4.4	3.5	2.8	2.4
	Group Comparison	t=0.94, p=.346, d=.07		t=−1.24, p=.217, d=.11		t=0.36, p=.716, d=.00	

Note. For the gender comparison, the test for equality of variance was significant and Welch's t-test was used (Welch, 1938).

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